

SkillSwap Platform

Complete Feature Documentation

Project Overview

Project Name: SkillSwap

Description: A peer-to-peer skill exchange platform where users can teach and learn skills using a token-based economy.

Technology Stack: React.js, Node.js, Express.js, MongoDB, Socket.io, TailwindCSS

1. Authentication & User Management

1.1 Email/Password Authentication

- **User Registration:** Sign up with name, email, password, and role selection (learner/teacher/both)
- **User Login:** Traditional email and password authentication
- **Email Activation:** Account activation via secure email link (required before first login)
- **Resend Activation Email:** Request new activation link if the original expired

1.2 OTP (One-Time Password) Login

- **Request OTP:** Send 6-digit verification code to registered email
- **Verify OTP:** Passwordless login using email verification code
- **Security Features:** 10-minute OTP expiry, maximum 5 verification attempts
- **Brute Force Protection:** Rate limiting on OTP requests

1.3 Password Management

- **Forgot Password:** Request password reset via email
- **Reset Password:** Set new password using secure reset link

1.4 Profile Management

- **View Profile:** Display user details, skills, and statistics
- **Update Profile:** Edit name, bio, country, timezone, languages, and avatar
- **User Preferences:** Customize notification settings and theme preferences

2. Skills System

2.1 Skill Management

- **Browse Skills:** View all available skills organized by categories
- **Search Skills:** Find specific skills using keyword search
- **Skill Categories:** Technology, Languages, Arts & Crafts, Business, Music, Sports, Academics, Lifestyle
- **Add Skills to Teach:** Teachers can list skills they want to offer
- **Add Skills to Learn:** Learners can list skills they want to acquire

2.2 Skill Discovery

- **Find Teachers by Skill:** Search for users who teach specific skills
- **Skill Proficiency Levels:** Beginner, Intermediate, Advanced, Expert
- **Hourly Token Rates:** Teachers set their rates per hour of instruction

3. Token Economy System

3.1 Token Management

- **Welcome Bonus:** 50 free tokens awarded upon registration
- **Token Balance:** Real-time tracking of current token balance
- **Tokens Earned:** Accumulated from teaching sessions
- **Tokens Spent:** Deducted for learning sessions

3.2 Transaction System

- **Transaction History:** Complete log of all token movements
- **Transaction Types:** Credit and Debit entries with detailed reasons
- **Balance Tracking:** Before and after balance recorded for each transaction

4. User Discovery & Social Features

4.1 Search & Browse

- **Search Users:** Find users by name or keyword
- **Browse Teachers:** View complete list of available teachers
- **Filter by Skill:** Find users teaching specific skills
- **User Profiles:** Detailed teacher and learner profile pages

4.2 Social Interaction

- **Follow Users:** Follow favorite teachers and learners
- **Followers/Following Lists:** View and manage social connections
- **User Ratings:** Average rating and total review count display
- **Teacher Statistics:** Sessions taught, experience level, and achievements

5. Real-Time Messaging System

5.1 Chat Features

- **Direct Messages:** Send private messages to any user
- **Conversations List:** View all active conversations
- **Real-Time Updates:** Instant message delivery using Socket.io
- **Message History:** Access and scroll through previous messages
- **Read Receipts:** Track message read status

6. Session Booking System

- **Book Sessions:** Schedule learning sessions with teachers
- **Session Status Tracking:** Pending, Confirmed, Completed, Cancelled states
- **Token Transfer:** Automatic token exchange upon session completion

7. Gamification Features

7.1 Streak System

- **Daily Streak:** Track consecutive days of platform activity
- **Current Streak Counter:** Display ongoing streak count
- **Longest Streak Record:** Personal best streak achievement
- **Last Activity Tracking:** Monitor user engagement

7.2 Leveling System

- **User Levels:** Progressive levels based on experience points
- **Experience Points:** Earned through various platform activities
- **Achievement Badges:** Special badges for milestones and accomplishments

8. Admin Features

- **User Management:** View, edit, and manage all platform users
- **Role Management:** Assign and modify user roles (User, Teacher, Admin)
- **Platform Statistics:** Overview of platform metrics and activity

9. Security Features

9.1 Rate Limiting

- **Authentication Rate Limit:** 5 attempts per 15 minutes for login/register
- **OTP Rate Limit:** 3 OTP requests per 15 minutes
- **General API Rate Limit:** 100 requests per 15 minutes

9.2 Input Validation & Authentication

- **Email Validation:** Proper email format verification
- **Password Strength:** Minimum security requirements enforced
- **Data Sanitization:** XSS and SQL injection protection
- **JWT Tokens:** Secure JSON Web Tokens with 7-day expiry
- **Protected Routes:** Authentication required for sensitive endpoints

10. Frontend Pages

Page Name	Description
Landing Page	Welcome page for new visitors
Sign Up	User registration form
Sign In	User login form
Dashboard	User home page after authentication
Home	Main browsing and discovery page
Profile	User profile view and edit page
Teacher Profile	Detailed teacher information page
Skill Search Results	Search results display page
Activate Account	Email activation confirmation page
OTP Login	OTP request page
OTP Verify	OTP code entry page
Admin Dashboard	Administrative control panel

11. Technical Architecture

11.1 Backend Stack

- **Runtime:** Node.js
- **Framework:** Express.js
- **Database:** MongoDB with Mongoose ODM
- **Real-Time:** Socket.io for WebSocket connections
- **Email Service:** Nodemailer with Gmail SMTP

11.2 Frontend Stack

- **Framework:** React.js
- **Build Tool:** Vite

- **Styling:** TailwindCSS
- **State Management:** React Context API
- **HTTP Client:** Axios

11.3 Security Implementation

- **Password Hashing:** Bcrypt.js
- **Authentication:** JSON Web Tokens (JWT)
- **CORS:** Cross-Origin Resource Sharing configuration
- **Helmet:** HTTP security headers

Document Information

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